

THESSALONIKI MAKEDONIA, Greece

WMO: 166220

Lat: 40.520N

Lon: 22.971E

Elev: 22

StdP: 14.68

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.5	29.8	14.2	11.3	32.5	17.8	13.5	34.7	32.9	43.9	28.5	41.4	7.9	110	0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	94.9	71.0	91.8	70.7	89.8	70.1	74.8	88.0	73.4	86.7	72.3	85.3	10.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.2	111.1	82.9	68.6	105.3	81.5	67.7	102.0	80.7	38.3	88.1	37.0	86.9	35.9	85.3	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.9	20.6	17.6	DB	22.0	100.8	3.6	3.7	19.4	103.4	17.3	105.6	15.3	107.6	12.7	110.3
			WB	20.0	77.9	3.4	2.0	17.5	79.3	15.5	80.5	13.6	81.6	11.1	83.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	60.9	42.4	45.1	50.2	57.5	67.1	75.8	80.1	79.9	71.6	62.2	53.2	45.0	(6)
(7)		DBStd	14.39	6.17	6.12	5.68	5.09	4.91	4.75	3.68	3.55	4.67	5.63	6.47	6.64	(7)
(8)		HDD50	696	244	154	67	6	0	0	0	0	0	2	40	183	(8)
(9)		HDD65	3080	701	556	459	229	35	0	0	0	6	117	355	622	(9)
(10)		CDD50	4690	8	18	74	232	532	774	932	928	648	380	137	27	(10)
(11)		CDD65	1596	0	0	0	4	101	324	467	463	204	31	2	0	(11)
(12)		CDH74	14976	0	0	5	54	728	2944	5015	4756	1345	126	3	0	(12)
(13)		CDH80	5977	0	0	2	6	150	1141	2225	2071	369	13	0	0	(13)
(14)	Wind	WSAvg	6.5	6.5	6.6	6.8	6.2	6.3	7.2	7.4	6.8	6.5	5.7	5.8	6.4	(14)
(15)	Precipitation	PrecAvg	17.6	1.5	1.4	1.6	1.3	1.7	1.1	0.9	0.8	1.3	1.6	2.0	2.2	(15)
(16)		PrecMax	28.9	4.4	6.6	4.2	4.3	3.9	3.4	3.7	3.9	5.6	7.6	7.0	6.5	(16)
(17)		PrecMin	9.3	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0	(17)
(18)		PrecStd	3.8	1.0	1.2	1.0	1.0	1.0	0.9	0.9	0.7	1.3	1.5	1.4	1.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.6	64.7	71.8	79.7	87.7	96.6	98.9	98.2	91.3	82.0	73.2	66.6	(19)
(20)		MCWB	53.3	52.9	55.6	60.6	65.4	70.6	71.0	70.8	68.1	67.1	61.2	57.3	(20)	
(21)		2%	DB	57.5	60.9	66.6	74.9	84.0	92.0	95.0	94.7	87.5	77.2	68.4	62.2	(21)
(22)		MCWB	50.7	51.6	53.8	59.4	64.7	70.3	70.9	71.4	67.8	64.5	60.3	54.9	(22)	
(23)		5%	DB	54.0	57.6	63.5	71.4	80.9	89.5	92.1	91.7	84.3	74.4	66.0	57.5	(23)
(24)		MCWB	48.7	50.4	52.6	57.4	63.7	69.3	70.8	71.1	67.3	63.1	59.0	52.3	(24)	
(25)		10%	DB	51.6	55.1	60.5	68.0	78.5	86.4	89.7	89.6	81.0	71.7	62.9	55.0	(25)
(26)		MCWB	47.3	48.9	51.8	56.4	63.0	68.8	70.3	70.6	66.9	62.3	57.2	50.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.6	55.4	58.8	63.7	69.6	75.5	76.5	77.2	72.6	69.4	63.8	58.9	(27)
(28)		MCDWB	60.4	61.3	65.8	73.6	82.2	88.4	90.5	90.0	83.3	77.5	68.8	66.2	(28)	
(29)		2%	WB	51.9	53.1	55.9	61.1	67.4	73.1	74.6	75.1	70.8	66.9	61.6	55.8	(29)
(30)		MCDWB	55.7	58.7	62.9	70.5	78.8	86.5	88.2	88.1	81.1	74.0	66.7	60.4	(30)	
(31)		5%	WB	49.9	51.3	54.2	59.4	65.8	71.7	73.2	73.6	69.5	65.1	59.8	53.3	(31)
(32)		MCDWB	53.0	56.5	61.1	67.8	76.4	84.6	86.6	86.9	79.8	71.6	64.9	57.0	(32)	
(33)		10%	WB	48.0	49.8	52.5	57.8	64.5	70.0	72.1	72.4	68.3	63.5	58.1	51.2	(33)
(34)		MCDWB	50.7	54.0	59.1	65.5	74.7	83.1	85.4	85.3	78.4	69.4	62.5	54.3	(34)	
(35)	Mean Daily Temperature Range	MDBR	13.0	14.3	16.7	18.4	19.8	20.0	20.1	19.8	18.1	15.7	14.0	12.8	(35)	
(36)		5% DB	MCDBR	16.9	19.2	22.8	24.3	23.9	23.6	23.5	22.9	22.4	20.1	16.4	15.8	(36)
(37)		MCWBR	12.1	12.5	13.1	12.5	11.0	10.5	11.0	10.9	11.1	11.2	10.7	10.9	(37)	
(38)		5% WB	MCDBR	14.4	16.9	20.1	20.8	21.4	21.8	21.5	21.1	20.4	16.9	14.3	12.8	(38)
(39)		MCWBR	10.9	11.6	12.6	12.0	10.6	10.5	10.5	10.6	10.9	10.4	9.9	9.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.359	0.390	0.433	0.477	0.465	0.459	0.459	0.471	0.451	0.425	0.392	0.368	(40)	
(41)		taud	2.363	2.242	2.130	2.030	2.139	2.199	2.204	2.164	2.193	2.237	2.279	2.325	(41)	
(42)		Ebn at Noon	249	256	257	254	260	261	260	253	249	240	232	233	(42)	
(43)		Edh at Noon	28	36	44	53	48	45	45	45	42	36	30	27	(43)	
(44)	All-Sky Solar Radiation	RadAvg	572	826	1202	1589	1942	2200	2267	2013	1483	993	655	511	(44)	
(45)		RadStd	83	105	121	166	95	128	93	94	108	117	76	84	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	-184	N/A	N/A	(46)
(47)	Regional (17 neighbors)	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	-205	N/A	N/A	(47)

Nomenclature: See separate page